Q.2

i) $f(x) = \begin{cases} \dfrac{3}{K^2}, & 0 < x < K \\ 0, & \text{otherwise} \end{cases}$

MGF $M_X(t) = \int_0^K e^{tx} \dfrac{3}{K^2} dx$

$= \dfrac{3}{K^2} \int_0^K e^{tx} dx = \dfrac{3}{K^2} \cdot \dfrac{e^{tx}}{t} \Big|_0^K$

$= \dfrac{3}{K^2} \left[\dfrac{e^{tK} - 1}{t} \right]$

$= \dfrac{3}{tK^2} \left[1 + tK + \dfrac{t^2K^2}{2!} + \dfrac{t^3K^3}{3!} + \dots + \dots - 1 \right]$

$= \dfrac{3}{tK^2} \left[tK + \dfrac{t^2K^2}{2!} + \dfrac{t^3K^3}{3!} + \dots \right]$ —— (2)

$E(x) = \dfrac{d}{dt} M_X(t) \Big|_{t=0}$

$= \dfrac{d}{dt} \left[\dfrac{3}{K^2} \left(K + \dfrac{tK^2}{2!} + \dfrac{t^2K^3}{3!} + \dots \right) \right]_{t=0}$

$= \dfrac{3}{K^2} \left(0 + \dfrac{K^2}{2!} + \cancel{\dfrac{2tK^3}{3!}} \to 0 + \dots \right) \Big|_{t=0}$

$= \dfrac{3}{K^2} \left(\dfrac{K^2}{2!} \right)$

$= \dfrac{3}{2}$

$E(x^2) = \dfrac{d^2}{dt^2} M_X(t) \Big|_{t=0}$

$= \dfrac{d}{dt} \left[\dfrac{3}{K^2} \left(\dfrac{K^2}{2!} + \dfrac{2tK^3}{3!} + \dfrac{3t^2K^4}{4!} + \dots \right) \right]_{t=0}$

$= \dfrac{3}{K^2} \left(0 + \dfrac{2K^3}{3!} + \dfrac{6tK^4}{4!} + \dots \right) \Big|_{t=0}$

$= \dfrac{3}{K^2} \left(\dfrac{2K^3}{3!} \right)$

$= K$

$V(x) = E(x^2) - [E(x)]^2$

$= K - \left(\dfrac{3}{2} \right)^2$

$= K - \dfrac{9}{4}$ —— (2)

ii) $f(x,y) = \begin{cases} 6x, & 0<x<2, 0<y<(1-x) \\ 0 & \text{otherwise} \end{cases}$

→ Two random variables x and Y are independent if

$$\boxed{f(x,y) = f_X(x) \cdot f_Y(y)}$$

Marginal density of X is $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

$$= \int_0^{1-x} 6x\,dy = 6x\, y\Big|_0^{1-x}$$

$$= \begin{cases} 6x(1-x) & , 0<x<2 \\ 0 & \text{otherwise} \end{cases}$$

Marginal density of Y is $f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

$$= \int_0^2 6x\,dx = 6 \cdot \frac{x^2}{2}\Big|_0^2 = \begin{cases} 12, & 0<y<1-x \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) \cdot f_Y(y) = 6x(1-x) \cdot 12$$
$$= 72x(1-x) \neq f(x,y)$$

∴ X and Y are not statistically independent. — (2)

→ $P(X>0.3 \mid Y=0.5) = \int_{0.3}^{2} f(x|y)\,dx$

$$f(x|y) = \frac{f(x,y)}{f(y)} = \frac{6x}{12} = \frac{x}{2}$$

$$P(X>0.3 \mid Y=0.5) = \int_{0.3}^{2} \frac{x}{2}\Big|_{y=0.5} dx = \frac{1}{2}\left|\frac{x^2}{2}\right|_{0.3}^{2}$$

$$= \frac{1}{4}[4-0.09] = 0.9775$$ — (2)

$E(xy) = 3.2$
$E(x) = -8$
$E(y) = 6$

(iii) A fair coin is tossed 1000 times.

In case of flipping a fair coin, the distⁿ of the number of heads follows a binomial distⁿ.

For fair coin, $p = 0.5$, $n = 1000$

$$\mu = 1000 \times 0.5 = 500$$

$$\sigma = \sqrt{1000 \times 0.5 \times 0.5} = 15.81$$

Let x = no. of heads

For CLT $P(480 \le X \le 520)$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z_{480} = \frac{480 - 500}{15.81} \approx -1.26$$

$$Z_{520} = \frac{520 - 500}{15.81} \approx 1.26$$

$$P(-1.26 \le z \le 1.26)$$

$$= 2P(0 \le Z < 1.26)$$

$$= \boxed{0.778}$$ (2) $\boxed{0.796}$

For Chebyshev's inequality:

$$K = \pm 1.27$$

$$P(|x - \mu| \le k\sigma) \ge 1 - \frac{1}{k^2}$$ (1)

Minimum portion of the data lie within k standard deviation of mean:

$$\left(1 - \frac{1}{k^2}\right) \times 100\% = \left(1 - \frac{1}{1.27^2}\right) \times 100\% = 37.50\% \approx \boxed{0.38}$$

Upper bound ≈ 0.62

Comparing the results CLT provides more accurate estimate of the probability in the given range.

(1)